

- **Result 1:** $\mathbf{x}$ is a **BFS** of $A_{m \times n} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \Leftrightarrow \mathbf{x}$ is an **extreme point** of $S_0 = \{\mathbf{x} \in \mathbb{R}^n : A_{m \times n} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$.

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Let $S' = \{\mathbf{x} \in \mathbb{R}^n : A_{m \times n} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$

and $S = \left\{ \begin{bmatrix} \mathbf{x} \\ \mathbf{s} \end{bmatrix} \in \mathbb{R}^{n+m} : A_{m \times n} \mathbf{x} + \mathbf{s} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}, \mathbf{s} \geq \mathbf{0} \right\}.$

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- **Result 2:** $\begin{bmatrix} \mathbf{x} \\ \mathbf{s} \end{bmatrix}$ is an **extreme point** of $S \Leftrightarrow \mathbf{x}$ is an **extreme point** of S' .

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Result 3: If $S \neq \emptyset$, then it has **at least one** BFS.

- **Result 4:** If $S_0 = \{\mathbf{x} \in \mathbb{R}^n : A_{m \times n} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\} \neq \phi$, then it has atleast one **BFS**.

- **Result 4:** If $S_0 = \{\mathbf{x} \in \mathbb{R}^n : A_{m \times n} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\} \neq \phi$, then it has atleast one **BFS**.
- **Result 5:** $\mathbf{d}_0$ is an **extreme direction** of $S' \Leftrightarrow \begin{bmatrix} \mathbf{d}_0 \\ \mathbf{d}'_0 \end{bmatrix}$ is an **extreme direction** of S where $\mathbf{d}'_0 \geq \mathbf{0}$ is such that $A\mathbf{d}_0 + \mathbf{d}'_0 = \mathbf{0}$.

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- In simplex method in any iteration we move from one **extreme point** to an **adjacent extreme point** if the current BFS is **nondegenerate**.

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- In simplex method in any iteration we move from one **extreme point** to an **adjacent extreme point** if the current BFS is **nondegenerate**.
- It may remain at the **same extreme point** if the current BFS is **degenerate**.

Sensitivity Analysis:

- Consider the problem **(P)**,

$$\text{Min } \mathbf{c}^T \mathbf{x}$$

subject to

$$A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}.$$

Let $\mathbf{x}_0$ be an **optimal solution** of this problem. WLOG let

$B = [\tilde{\mathbf{a}}_1, \dots, \tilde{\mathbf{a}}_m]$ be a basis matrix corresponding to $\mathbf{x}_0$,

hence a set of **basic variables** of $\mathbf{x}_0$ are $x_1, \dots, x_m$.

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- Changing the cost vector c:**
- If the new $c'_j - z'_j \geq 0$ for all $j = 1, \dots, n$, then $\mathbf{x}_0$ will again be **optimal** for the new problem.

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- Changing the cost vector c:**
- If the new $c'_j - z'_j \geq 0$ for all $j = 1, \dots, n$, then $\mathbf{x}_0$ will again be **optimal** for the new problem.
- If not, then **simplex algorithm** can be used to get an **optimal solution** for the new problem or to conclude that the new problem has **no optimal solution**.

Changing the vector $\mathbf{b}$:

- If the vector $\mathbf{b}$ is changed to $\mathbf{b}'$, and if the basic solution $\mathbf{x}'_0 = [(B^{-1}\mathbf{b}')^T, \mathbf{0}_{1 \times (n-m)}]^T$ corresponding to $\mathbf{x}_0 = [(B^{-1}\mathbf{b})^T, \mathbf{0}_{1 \times (n-m)}]^T$, is such that $B^{-1}\mathbf{b}' \geq \mathbf{0}$, then $\mathbf{x}'_0$ is optimal for the new problem.

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- Since $\mathbf{y}^T$ satisfies $\mathbf{y}^T \tilde{\mathbf{a}}_j = z_j = c_j$ for $j = 1, \dots, m$, it lies on m LI hyperplanes defining $\text{Fea}(D)$, so $\mathbf{y}$ is an extreme point of $\text{Fea}(D)$.

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- $\mathbf{y}^T = \mathbf{c}_B^T B^{-1}$ belongs to $\text{Fea}(D)$ (since $c_j - z_j \geq 0$, for all $j = 1, \dots, n$).
- Since $\mathbf{y}^T$ satisfies $\mathbf{y}^T \tilde{\mathbf{a}}_j = z_j = c_j$ for $j = 1, \dots, m$, it lies on **m LI hyperplanes** defining $\text{Fea}(D)$, so $\mathbf{y}$ is an **extreme point** of $\text{Fea}(D)$.
- The **dual simplex algorithm** can be used to either get an **optimal solution** of the new problem or to conclude that the new problem **does not** have a **feasible solution**.

- **Example :** Consider the LPP given by

$$\text{Max } -x_1 + 2x_2$$

subject to

$$-x_1 + x_2 \leq 1,$$

$$x_1 + x_2 \leq 7,$$

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If we convert the above problem to a problem with equality constraints by adding **(slack)** variables, then it becomes

- **Max** $-x_1 + 2x_2$

subject to

$$-x_1 + x_2 + s_1 = 1,$$

$$x_1 + x_2 + s_2 = 7,$$

$$x_1 + 3x_2 + s_3 = 15,$$

$$x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, s_2 \geq 0, s_3 \geq 0.$$

- The **optimal BFS** $[3, 4, 0, 0, 0]^T$ is **degenerate** and corresponds to three different basis matrix, $[\tilde{\mathbf{a}}_1, \tilde{\mathbf{a}}_2, \mathbf{e}_1]$, $[\tilde{\mathbf{a}}_1, \tilde{\mathbf{a}}_2, \mathbf{e}_2]$ and $[\tilde{\mathbf{a}}_1, \tilde{\mathbf{a}}_2, \mathbf{e}_3]$.
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The tables corresponding to these three bases are given by



$c_j - z_j$	0	0	0	$\frac{5}{2}$	$-\frac{3}{2}$	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$			0	$\frac{3}{2}$	$-\frac{1}{2}$	3
$\tilde{\mathbf{a}}_2$			0	$-\frac{1}{2}$	$\frac{1}{2}$	4
$\mathbf{s}_1$			1	2	-1	0

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- Note that **not** all the $c_j - z_j$ values in the above table are **nonpositive**, but the above BFS is still **optimal**.

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$\tilde{\mathbf{a}}_1$			0	$\frac{3}{2}$	$-\frac{1}{2}$	3
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$\mathbf{s}_1$			1	2	-1	0

- Note that **not** all the $c_j - z_j$ values in the above table are **nonpositive**, but the above BFS is still **optimal**.
- So the optimality condition, $c_j - z_j \geq 0$ for all j , is a **sufficient condition** but **not** a **necessary condition** for the corresponding BFS to be optimal.



$c_j - z_j$	0	0	$-\frac{5}{4}$	0	$-\frac{1}{4}$	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$			$-\frac{3}{4}$	0	$\frac{1}{4}$	3
$\tilde{\mathbf{a}}_2$			$\frac{1}{4}$	0	$\frac{1}{4}$	4
$\mathbf{s}_2$			$\frac{1}{2}$	1	$-\frac{1}{2}$	0

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$\tilde{\mathbf{a}}_1$			$-\frac{3}{4}$	0	$\frac{1}{4}$	3
$\tilde{\mathbf{a}}_2$			$\frac{1}{4}$	0	$\frac{1}{4}$	4
$\mathbf{s}_2$			$\frac{1}{2}$	1	$-\frac{1}{2}$	0

$c_j - z_j$	0	0	$-\frac{3}{2}$	$-\frac{1}{2}$	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$			$-\frac{1}{2}$	$\frac{1}{2}$	0	3
$\tilde{\mathbf{a}}_2$			$\frac{1}{2}$	$\frac{1}{2}$	0	4
$\mathbf{s}_3$			-1	-2	1	0

- The dual of the above problem is given by

$$\text{Min } y_1 + 7y_2 + 15y_3$$

subject to

$$-y_1 + y_2 + y_3 \geq -1,$$

$$y_1 + y_2 + 3y_3 \geq 2,$$

$$y_1 \geq 0, y_2 \geq 0, y_3 \geq 0.$$

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- The **optimal solutions** of the Dual obtained from the optimal tables are given by:

$$\mathbf{y}^T = \mathbf{c}_B^T B^{-1} = \mathbf{c}_B^T B^{-1} [\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3] = [z_{s_1}, z_{s_2}, z_{s_3}]$$

where z_{s_i} is the z_j value corresponding to the **slack variable** s_i .

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where z_{s_i} is the z_j value corresponding to the **slack variable** s_i .
- $[\frac{5}{4}, 0, \frac{1}{4}]^T$ and $[\frac{3}{2}, \frac{1}{2}, 0]^T$, are both **optimal solutions** of the Dual as well as **extreme points** of $\text{Fea}(D)$.

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- $[\frac{5}{4}, 0, \frac{1}{4}]^T$ and $[\frac{3}{2}, \frac{1}{2}, 0]^T$, are both **optimal solutions** of the Dual as well as **extreme points** of $\text{Fea}(D)$.
- The dual has **infinitely many** optimal solutions.

- If we convert the dual problem into a problem with equality constraints by adding (surplus) variables then we get

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- Min $y_1 + 7y_2 + 15y_3$
subject to
$$\begin{aligned} -y_1 + y_2 + y_3 - s'_1 &= 1, \\ y_1 + y_2 + 3y_3 - s'_2 &= 2, \\ y_1 \geq 0, y_2 \geq 0, y_3 \geq 0, s'_1 \geq 0, s'_2 \geq 0. \end{aligned}$$

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- The BFS corresponding to the extreme point $[\frac{3}{2}, \frac{1}{2}, 0]^T$ have basic variables as y_1, y_2 .

- The table corresponding to these basic feasible solutions will be given by (check this)

$c_j - z_j$	0	0	0	3	4	
	$B^{-1}\tilde{\mathbf{a}}'_1$	$B^{-1}\tilde{\mathbf{a}}'_2$	$B^{-1}\tilde{\mathbf{a}}'_3$	$B^{-1}(-\mathbf{e}_1)$	$B^{-1}(-\mathbf{e}_2)$	B^{-1}
$\tilde{\mathbf{a}}'_1$					$-\frac{1}{2}$	$\frac{3}{2}$
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	$B^{-1}\tilde{\mathbf{a}}'_1$	$B^{-1}\tilde{\mathbf{a}}'_2$	$B^{-1}\tilde{\mathbf{a}}'_3$	$B^{-1}(-\mathbf{e}_1)$	$B^{-1}(-\mathbf{e}_2)$	B^{-1}
$\tilde{\mathbf{a}}'_1$					$-\frac{1}{4}$	$\frac{5}{4}$
$\tilde{\mathbf{a}}'_3$					$-\frac{1}{4}$	$\frac{1}{4}$

- The table corresponding to these basic feasible solutions will be given by (check this)

$c_j - z_j$	0	0	0	3	4	
	$B^{-1}\tilde{\mathbf{a}}'_1$	$B^{-1}\tilde{\mathbf{a}}'_2$	$B^{-1}\tilde{\mathbf{a}}'_3$	$B^{-1}(-\mathbf{e}_1)$	$B^{-1}(-\mathbf{e}_2)$	B^{-1}
$\tilde{\mathbf{a}}'_1$					$-\frac{1}{2}$	$\frac{3}{2}$
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	$B^{-1}\tilde{\mathbf{a}}'_1$	$B^{-1}\tilde{\mathbf{a}}'_2$	$B^{-1}\tilde{\mathbf{a}}'_3$	$B^{-1}(-\mathbf{e}_1)$	$B^{-1}(-\mathbf{e}_2)$	B^{-1}
$\tilde{\mathbf{a}}'_1$					$-\frac{1}{4}$	$\frac{5}{4}$
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Here $\tilde{\mathbf{a}}'_i$ gives the columns corresponding to the variables $y_i, i = 1, 2, 3$ in the dual constraints, when the constraints are written in the greater than equal to form.

The primal optimal table and the corresponding dual table:

$c_j - z_j$	0	0	$-\frac{5}{4}$	0	$-\frac{1}{4}$	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$			$-\frac{3}{4}$	0	$\frac{1}{4}$	3
$\tilde{\mathbf{a}}_2$			$\frac{1}{4}$	0	$\frac{1}{4}$	4
$\mathbf{s}_2$			$\frac{1}{2}$	1	$-\frac{1}{2}$	0

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$\tilde{\mathbf{a}}_1$			$-\frac{3}{4}$	0	$\frac{1}{4}$	3
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$c_j - z_j$	0	0	0	3	4	
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$c_j - z_j$	0	0	$-\frac{3}{2}$	$-\frac{1}{2}$	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$			$-\frac{1}{2}$	$\frac{1}{2}$	0	3
$\tilde{\mathbf{a}}_2$			$\frac{1}{2}$	$\frac{1}{2}$	0	4
$\mathbf{s}_3$			-1	-2	1	0

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$\tilde{\mathbf{a}}_1$			$-\frac{1}{2}$	$\frac{1}{2}$	0	3
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- The new problem is given as following:

$$\text{Max } -x_1 + 2x_2$$

subject to

$$-x_1 + x_2 + s_1 = 1,$$

$$x_1 + x_2 + s_2 = 7,$$

$$x_1 + 3x_2 + s_3 = 14,$$

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- Note that the BFS corresponding to the basis $[\tilde{a}_1, \tilde{a}_2, e_1]$, should have $s_2 = s_3 = 0$ hence should lie at the intersection of the two lines

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- The corresponding BFS is $[\frac{7}{2}, \frac{7}{2}, 1, 0, 0]^T$, which is a **nondegenerate** BFS.

- Since all the $c_j - z_j$ values are not ≤ 0 , so by entering s_2 into the basis one can get a BFS with better value of the objective function.
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- The corresponding BFS is $[\frac{11}{4}, \frac{15}{4}, 0, \frac{1}{2}, 0]^T$, a **nondegenerate** BFS.
- Since $c_j - z_j \leq 0$ for all $j = 1, \dots, n$, so this BFS is **optimal solution** for the new problem.

- The **basic solution** corresponding to the basis $[\tilde{\mathbf{a}}_1, \tilde{\mathbf{a}}_2, \mathbf{e}_3]$, should have $s_1 = s_2 = 0$ hence should lie at the intersection of the two lines
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- The corresponding basic solution is $[3, 4, 0, 0, -1]^T$ which is not feasible for the new problem.
- Will the new dual now have a unique optimal solution or will it again have infinitely many optimal solutions?

Dual Simplex Algorithm :

- Consider the following LP problem (P) :
Min $\mathbf{c}^T \mathbf{x}$
subject to $A_{m \times n} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$.

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- In this case the **Dual Simplex Method** is used to get an **optimal solution** of (P) or to conclude that (P) **does not** have a **feasible solution**.

- Take x_r to be the **leaving basic variable** if $(B^{-1}\mathbf{b})_r = \min\{(B^{-1}\mathbf{b})_j : (B^{-1}\mathbf{b})_j < 0\}$.

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- Hence the primal problem (P) **does not** have a **feasible solution**.

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- Then x_s is the **entering variable** if

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So the **pivot element** is u_{rs} .

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 So the **pivot element** is u_{rs} .
- The table is updated by performing the necessary elementary row operations.
- The new column corresponding to $\tilde{\mathbf{a}}_s$ in the table is the r th column of I_m and new $c_s - z'_s = 0$.
- New $c_j - z'_j = c_j - z_j - \frac{c_s - z_s}{u_{rs}} u_{rj}$.

- If $\mathbf{x}'$ is the **new basic solution** of the primal (P) and $\mathbf{y}'$ be the corresponding **feasible solution** of the Dual, then
 $\mathbf{b}^T \mathbf{y}' = \mathbf{c}^T \mathbf{x}' = \mathbf{c}^T \mathbf{x} + (c_s - z_s) \frac{x_r}{u_{rs}} \geq \mathbf{c}^T \mathbf{x} = \mathbf{b}^T \mathbf{y}$.

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- Since the Dual is a **maximization problem** so $\mathbf{y}'$ is a **better solution** with respect to cost (or the objective function) than $\mathbf{y}$.
- If the new **basic solution** $\mathbf{x}'$ is non negative, then $\mathbf{x}'$ is an **optimal BFS** of (P).
- If not repeat the procedure till you get a **BFS** and hence an **optimal solution** of (P) or conclude that (P) has **no feasible solution**.

- In case (P) is **Max** $\mathbf{c}^T \mathbf{x}$
 subject to $\mathbf{A}_{m \times n} \mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$.
 Then Dual of (P) is given by
Min $\mathbf{b}^T \mathbf{y}$
 subject to $\mathbf{A}_{n \times m}^T \mathbf{y} \geq \mathbf{c}$.
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- **Case 1'**: $u_{rj} \geq 0$ for all $j = 1, 2, \dots, n$.
Then the primal (P) **does not** have a **feasible solution**.
- $(B^{-1})_r$ is a **direction** for $\text{Fea}(D)$.
- If we denote $(B^{-1})_r$ as $\mathbf{d}_0^T$,
 $(B^{-1} \mathbf{b})_r = \mathbf{d}_0^T \mathbf{b} < 0$.
- Since the **Dual** is a **minimization problem** it follows that the Dual **does not** have an **optimal solution**.

- In case (P) is **Max** $\mathbf{c}^T \mathbf{x}$
subject to $\mathbf{A}_{m \times n} \mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$.
Then Dual of (P) is given by
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subject to $\mathbf{A}_{n \times m}^T \mathbf{y} \geq \mathbf{c}$.
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- Hence the primal problem (P) **does not** have a **feasible solution**.

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$$\frac{c_s - z_s}{u_{rs}} = \min_j \left\{ \frac{c_j - z_j}{u_{rj}} : u_{rj} < 0 \right\}.$$

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- The table is updated by performing the necessary elementary row operations.
- The new column corresponding to $\tilde{\mathbf{a}}_s$ in the table is the r th column of I_m and $c_s - z'_s = 0$.

- If $\mathbf{x}'$ is the **new basic solution** of the primal (P) and $\mathbf{y}'$ be the corresponding **feasible solution** of the Dual, then
 $\mathbf{b}^T \mathbf{y}' = \mathbf{c}^T \mathbf{x}' = \mathbf{c}^T \mathbf{x} + (c_s - z_s) \frac{x_r}{u_{rs}} \leq \mathbf{c}^T \mathbf{x} = \mathbf{b}^T \mathbf{y}.$

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- Since the Dual is a **minimization problem** so $\mathbf{y}'$ is a **better solution** with respect to cost (or the objective function) than $\mathbf{y}$.
- If the new **basic solution** $\mathbf{x}'$ is non negative, then $\mathbf{x}'$ is an **optimal BFS** of (P).
- If not repeat the procedure till you get a **BFS** and hence an **optimal solution** of (P) or conclude that (P) has **no feasible solution**.

● Max $-3x_1 + 2x_2$

subject to

$$-x_1 + x_2 + s_1 = 1,$$

$$x_1 + x_2 + s_2 = 7,$$

$$x_1 + 3x_2 + s_3 = 15,$$

$$x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, s_2 \geq 0, s_3 \geq 0.$$

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The table corresponding to the basic solution with $x_1 = -1$ and $x_2 = 0$ is given by



$c_j - z_j$		-1	-3			
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$		-1	-1			-1
$\mathbf{s}_2$		2	1			8
$\mathbf{s}_3$		4	1			16

- According to the Dual Simplex method, x_1 is the leaving variable, since $(B^{-1}\mathbf{b})_1 < 0$ and $(B^{-1}\mathbf{b})_2 > 0$, $(B^{-1}\mathbf{b})_3 > 0$.

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- Also since in the row corresponding to the leaving variable, only two entries are negative, u_{12} , and u_{13} and $\frac{c_2 - z_2}{u_{12}} = 1 < \frac{c_3 - z_3}{u_{13}} = 3$, hence the entering variable is x_2 and the **pivot** is u_{12} , where u_{ij} 's have their usual meaning.

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- Also since in the row corresponding to the leaving variable, only two entries are negative, u_{12} , and u_{13} and $\frac{c_2 - z_2}{u_{12}} = 1 < \frac{c_3 - z_3}{u_{13}} = 3$, hence the entering variable is x_2 and the **pivot** is u_{12} , where u_{ij} 's have their usual meaning. After doing the necessary elementary row operations we get the following table :



$c_j - z_j$	-1	0	-2			
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_2$	-1	1	1			1
$\mathbf{s}_2$	2	0	-1			6
$\mathbf{s}_3$	4	0	-3			12

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	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_2$	-1	1	1			1
$\mathbf{s}_2$	2	0	-1			6
$\mathbf{s}_3$	4	0	-3			12

Hence an optimal solution of the primal is given by, $x_2 = 1$ and $x_1 = 0$.

The optimal solution of the Dual is given by,

$$y_1 = 2, y_2 = 0, y_3 = 0,$$

where the Dual is given by:

$$\text{Min } y_1 + 7y_2 + 15y_3$$

subject to

$$-y_1 + y_2 + y_3 \geq -3,$$

$$y_1 + y_2 + 3y_3 \geq 2,$$

$$y_1 \geq 0, y_2 \geq 0, y_3 \geq 0.$$